Diploma in Foundation of Artificial Intelligence & Data Science

Executive Summary

This document outlines a comprehensive, six-month (24 weeks – 600 hours), “Diploma in Foundations of Artificial Intelligence & Data Science (AI-DS)” designed for undergraduates and graduates. The programme directly addresses the massive industry demand for skilled AI/DS professionals, a gap evident in traditional engineering curricula. With a pedagogy blending 30% theory with 70% hands-on labs and real-life industry project-based learning, the curriculum covers foundational programming, data handling, classical machine learning, and advanced deep learning, including generative AI and MLOps. Aligned with the National Education Policy (NEP 2020) and India's National AI Mission, this diploma aims to produce industry-ready talent capable of securing roles like Data Analyst, ML Engineer, and AI Specialist. The curriculum is also mapped to the knowledge domains of leading international certifications from AWS, Google, and Microsoft, preparing graduates for global recognition. Delivered through a network of high-quality partner centres, this initiative will equip students with the practical skills and theoretical knowledge needed to thrive in the digital economy.

1. Background & Rationale

1.1 Market Demand for AI/DS Skills in India

The demand for professionals skilled in Artificial Intelligence (AI) and Data Science (DS) is experiencing exponential growth in India. A 2024 NASSCOM report highlights that the Indian AI market is projected to reach USD 17 billion by 2027, creating over 500,000 high-skill jobs. Similarly, data from platforms like LinkedIn and Naukri consistently show "Data Scientist" and "Machine Learning Engineer" among the top emerging jobs, with a persistent talent deficit of over 30% at the entry-level. Industries from finance (algorithmic trading, risk assessment) and healthcare (diagnostic imaging, drug discovery) to e-commerce (recommendation engines, supply chain optimization) and manufacturing (predictive maintenance, quality control) are aggressively adopting AI, driving the need for a workforce that can collect, analyse, model, and deploy data-driven solutions. This programme is strategically positioned to meet this burgeoning demand.

Exploring the Exponential Growth of AI/DS Skills in India

The booming demand for Artificial Intelligence and Data Science professionals in the Indian job market

$17billion

Projected growth of Indian AI market

The Indian AI market is expected to reach $17 billion by 2027.

500,000 jobs

High-skill job creation

This growth will create over 500,000 high-skill jobs in AI.

30% deficit

Talent shortage at entry level

There is a persistent talent deficit of over 30% in entry-level positions.

Top jobs

Emerging roles in AI/DS

'Data Scientist' and 'Machine Learning Engineer' are among the top emerging jobs.

Addressing the Gap in Engineering Education

Transforming theoretical knowledge into practical AI and Data Science expertise

1 Strengths

The diploma offers intensive, **hands-on training** in AI/DS, addressing industry demands.

2 Weaknesses

Limited **curriculum time** in engineering programs restricts depth of AI/DS training.

3 Opportunities

Growing **industry demand** for skilled professionals creates a need for specialized programs.

4 Threats

Rapid **technological changes** may outpace curriculum updates, affecting program relevance.

1.2 Gap in Typical Engineering Curricula

While many Indian engineering colleges have introduced AI/DS as elective subjects, these courses often remain theoretical, lacking the rigorous, hands-on, project-based training required by the industry and mostly is made available in the last year of engineering with limited time for students to grasp the concepts. A typical BE/BTech curriculum, constrained by its broad scope, cannot dedicate the 500+ intensive learning hours needed to develop practical proficiency in areas like model deployment (MLOps), large-scale data processing, and advanced deep learning applications. This diploma programme is designed specifically to bridge this gap, acting as a finishing school that transforms theoretical knowledge into job-ready practical expertise.

1.3 Alignment with National Education Policy (NEP 2020) & National AI Mission

The programme's philosophy is in direct alignment with key national strategic initiatives:

- **NEP 2020:** The policy emphasizes multidisciplinary learning, skill development, and a credit-based system. This diploma offers a modular, credit-alignable structure that can be undertaken by students from any engineering branch, promoting the interdisciplinary approach championed by NEP 2020. Its focus on practical problem-solving and "learning how to learn" are core tenets of the policy.
- **National AI Mission:** The Government of India's AI Mission aims to establish the country as a global leader in AI. A critical pillar of this mission is building a skilled workforce. This diploma directly contributes to this goal by creating a pipeline of trained professionals ready to contribute to the #AIforAll vision, fostering innovation and application of AI in key sectors.

2. Programme Snapshot

Parameter	Details
Programme Title	Diploma in Foundation of Artificial Intelligence & Data Science (AI-DS)
Duration	6 Months (24 weeks), Full-Time
Total Learning Hours	~600 Hours (Lectures, Labs, Projects), 5 hours per day, 5 days a week
Mode of Delivery	Physical Classroom / In-person at authorized lab supported with real time industry project focus with regular remote mentoring sessions
Target Audience	Under Graduates or recent graduates (any year, any branch)
Intake per Batch	60 Students (to maintain mentor-student ratio)
Eligibility Criteria	Basic programming logic is beneficial but not mandatory.

Certification	Certification by the MKCL upon successful completion.
Language(s)	English (Primary) & Marathi

3. Module-wise Detailed Syllabus

The programme is structured into four intensive modules, building from fundamentals to advanced specialisations.

3.1 Module 1: Foundations of AI, Data Science & Programming (Weeks 1–4, 120 Hours)

- **Indicative Software/Tools:** Python 3.11+, VS Code, Git, GitHub, Jupyter Notebooks, NumPy, Basic Linux Commands. Exposure to Freelancing Projects and Kaggle.

Weekly Breakdown:

Week	Lecture Themes	Lab Projects / Assignments	Key Learning Outcomes
1	Intro to AI/DS & AI Ethics, Python Fundamentals (Variables, Data Types, Control Flow)	Setup development environment. Write simple Python scripts. Git setup and first commit. Discuss a case study on algorithmic bias.	Understand the AI/DS landscape and its ethical implications. Write and execute basic Python programs. Use version control.
2	Python Data Structures (Lists, Tuples, Dictionaries, Sets), Functions & Modules	Develop a command-line contact book application. Create and import custom Python modules for utility functions.	Manipulate complex data structures. Write modular, reusable code.
3	Object-Oriented Programming (OOP) in Python, File Handling (CSV, JSON, TXT)	Design a simple inventory management system using classes and objects. Parse data from a JSON file and write results to a CSV.	Apply OOP principles. Perform data I/O operations with multiple file formats.
**4. Essential Maths for AI/DS (Linear Algebra, Probability, Statistics - an intuitive overview)	Use NumPy to perform vector/matrix operations. Calculate descriptive statistics on a sample dataset.	Grasp the core mathematical concepts underpinning ML algorithms. Use NumPy for numerical computation.	

This foundational module brings all students, especially those from non-Computer Science backgrounds, up to speed on essential programming and mathematical concepts crucial for AI and Data Science.

- **Topics Covered:**
  - **Python Programming Fundamentals:**
    - Basic syntax, data types (integers, floats, strings, booleans), operators.
    - Control structures (if/else, for loops, while loops).
    - Functions, modules, and packages.
    - **Object-Oriented Programming (OOP):** Classes, objects, inheritance, polymorphism, encapsulation, abstraction.
    - Error handling (try-except blocks).
  - **Command Line Interface (CLI) & Git Workflow:**
    - Basic shell commands (ls, cd, mkdir, rm).
    - Introduction to Git for version control: `init`, `add`, `commit`, `status`, `log`.
    - GitHub for collaborative development: `clone`, `push`, `pull`, `fork`, pull requests (PRs), merging.
  - **Environment Management:**
    - Understanding virtual environments (e.g., `virtualenv`, `conda`, `Poetry`).
    - Dependency management and `requirements.txt`.
  - **NumPy for Numerical Computing:**
    - Arrays (ndarrays), array creation, indexing, slicing.
    - Array operations (element-wise, broadcasting).
    - Linear algebra operations (dot product, matrix multiplication).
  - **Pandas for Data Manipulation:**
    - Series and DataFrames.
    - Data loading (CSV, Excel).
    - Data inspection (`head`, `info`, `describe`, `shape`).
    - Indexing, selecting, filtering data.
    - Grouping and aggregation (`groupby`).
    - Merging and joining DataFrames.
    - Handling missing values (`NaN`).
  - **Descriptive Statistics:** Measures of central tendency (mean, median, mode), measures of dispersion (variance, standard deviation, range, IQR).
  - **Inferential Statistics:**
    - Sampling, Central Limit Theorem.
    - Hypothesis testing (t-tests, chi-square tests, ANOVA).
    - P-values and confidence intervals.
  - **Probability Distributions:** Normal, Binomial, Poisson distributions. Bayes' Rule.
  - **Linear Algebra Fundamentals:**
    - Vectors and matrices: operations, norms.
    - Matrix decomposition (e.g., Eigenvalue decomposition, Singular Value Decomposition - SVD).
  - **Differential Calculus & Gradient Intuition:**
    - Derivatives, partial derivatives, chain rule.
    - Gradients and their role in optimization algorithms.
  - **Responsible AI Principles & Privacy Laws:**

        - Introduction to AI ethics, fairness, accountability, transparency.
        - Overview of data privacy regulations (GDPR, India DPDP Act).
- **Tools:** Python 3.12, VS Code, JupyterLab, Git/GitHub, NumPy, Pandas, SciPy, Matplotlib, virtualenv/Poetry.
- **Case Study:**
    - **COVID-19 Open-Data Audit:** Learners pull Johns Hopkins daily case files, clean, join, and plot 7-day moving averages. This involves handling time-series data, dealing with data inconsistencies, and discussing data quality and the risk of misinterpretation in public health reporting.
- **Real-life Example:**
    - **Johns Hopkins Dashboard as Global Pandemic "Single Source of Truth":** Emphasizes the importance of reproducible data pipelines, data governance, and clear visualization in public health crises. Discussion on the impact of data quality on public perception and policy.
- **Hands-on Activities:**
    - **Git-Immersion Lab:** A guided lab covering `git clone`, `git branch`, `git checkout`, `git add`, `git commit`, `git push`, creating a pull request on GitHub, and merging.
    - **Stats Worksheet on A/B Test Significance:** Apply statistical tests to determine the significance of A/B test results (e.g., website conversion rates).
    - **Final Mini-Project: Build a CLI Tool for CSV Summarization:** Develop a Python command-line tool using `argparse` and Pandas to take a CSV file as input and output basic statistics (counts, missing values, unique values, histograms for numerical columns, frequency counts for categorical columns).

3.2 Module 2: Data Handling & Visualisation (Weeks 5–8, 120 Hours)

- **Indicative Software/Tools:** Pandas, Matplotlib, Seaborn, Plotly, SQL (SQLite), Beautiful Soup. Industry Projects from Kaggle.

Weekly Breakdown:

Week	Lecture Themes	Lab Projects / Assignments	Key Learning Outcomes
5	Data Wrangling with Pandas: DataFrames, Series, Indexing, Selection, Cleaning	Load a real-world dataset (e.g., Titanic). Perform data cleaning: handle missing values, correct data types, detect outliers.	Load, clean, and preprocess tabular data efficiently using Pandas.
6	Data Aggregation & Transformation: GroupBy, Merging, Joining, Pivoting	Analyse a retail sales dataset. Aggregate sales by store/product. Merge customer data from another table to create a master view.	Transform and aggregate data to derive insights. Combine data from multiple sources.
7	Data Visualisation & EDA: Matplotlib, Seaborn, Intro to Interactive Plotting (Plotly)	Create a comprehensive Exploratory Data Analysis (EDA) report for a given dataset, including static and interactive plots.	Create meaningful visualisations to communicate data stories effectively.

8	Data Acquisition: SQL (Joins, Subqueries, Aggregations), Intro to Web Scraping	Write complex SQL queries to extract data from a relational database. Scrape product details from an e-commerce site.	Retrieve data from relational databases. Extract data from web pages/kaggle

This module focuses on effective data collection, storage, cleaning, and visualization techniques, preparing students to work with diverse data sources.

- **Topics Covered:**
  - **Data Ingestion Techniques:**
    - **REST APIs:** Making HTTP requests to fetch data from web services.
    - **Web Scraping:** Using libraries like Beautiful Soup and Scrapy to extract data from websites.
    - **File Formats:** Reading and writing data from various formats (CSV, JSON, XML, Parquet, ORC, Avro, Zip files).
  - **Data Cleaning Patterns:**
    - Handling missing values (imputation strategies: mean, median, mode, forward/backward fill, K-NN imputation).
    - Outlier detection and treatment (winsorisation, Z-score, IQR method).
    - Data type conversion and consistency checks.
    - Handling duplicate records.
  - **Feature Engineering Cheatsheet:**
    - Creating new features from existing ones (e.g., polynomial features, interaction terms).
    - Binning/discretization.
    - Log and square root transformations (log/x^2).
    - One-hot encoding, label encoding.
  - **Relational Databases (RDBMS):**
    - Entity-Relationship (ER) design.
    - SQL fundamentals: `SELECT`, `FROM`, `WHERE`, `GROUP BY`, `HAVING`, `ORDER BY`.
    - Joins (`INNER`, `LEFT`, `RIGHT`, `FULL`, `SELF`).
    - Subqueries and Common Table Expressions (CTEs).
    - **SQL Window Functions:** `ROW_NUMBER()`, `RANK()`, `LAG()`, `LEAD()`, aggregate window functions.
    - Database indexing for performance optimization.
  - **NoSQL Databases Introduction:**
    - Overview of different NoSQL types: Document (MongoDB), Key-Value, Columnar, Graph.
    - Use cases and advantages/disadvantages compared to RDBMS.
  - **Big Data Introduction:**
    - Concepts of distributed computing.
    - Overview of **Hadoop HDFS** (Hadoop Distributed File System).
    - Introduction to **Apache Spark** for in-memory distributed processing.
    - PySpark for Python integration with Spark.
  - **Exploratory Data Analysis (EDA) Workflow:**
    - Univariate, bivariate, and multivariate analysis.
    - Correlation matrices, scatter plots, box plots, histograms.
    - Identifying patterns, anomalies, and relationships.

- Design Principles of Dashboards:
  - Data visualization best practices (Edward Tufte's principles).
  - Choosing appropriate chart types.
  - Storytelling with data.
  - Common pitfalls (e.g., dark-theme pitfalls, misleading visualizations).
- **Tools:** MySQL 8, MongoDB 6, Apache Spark 3 (PySpark), Airbyte, Tableau Public, Power BI.
- **Case Studies:**
  - **Indian Retail Sales (Kaggle BigMart):** Perform extensive EDA on a retail sales dataset, identify key factors influencing sales, and design a sales dashboard to visualize trends and insights.
  - **Bengaluru AQI Sensor Feeds:** Work with real-time or historical Air Quality Index (AQI) time-series data from sensors, focusing on data cleaning, handling missing sensor readings (imputation), and preparing data for analysis.
- **Real-life Example:**
  - **Flipkart's Clickstream Warehouse (50 TB/day):** Explanation of how large e-commerce platforms handle massive volumes of clickstream data, underscoring why technologies like OLAP cubes (for analytical processing) and Spark (for distributed data processing) are essential for real-time analytics and business intelligence.
- **Hands-on Activities:**
  - **SQL Challenge Set:** Solve 20 complex SQL queries, including advanced joins, subqueries, and window functions, against a provided relational database schema.
  - **PySpark Notebook for AQI Aggregates:** Write a PySpark notebook to load, clean, and compute daily or weekly AQI aggregates from a large dataset, demonstrating distributed data processing.

Tableau Dashboard Design & Publishing: Design and publish an interactive Tableau dashboard based on a cleaned dataset. Students will peer-review each other's dashboards for clarity, storytelling, and adherence to design principles.

3.3 Module 3: Classical Machine Learning (Weeks 9–16, 240 Hours)

- **Indicative Software/Tools:** Scikit-learn, XGBoost, Statsmodels, MLflow.

Weekly Breakdown:

Week	Lecture Themes	Lab Projects / Assignments	Key Learning Outcomes
9	Intro to ML, Linear & Logistic Regression, Cost Functions, Gradient Descent	Build regression/classification models from scratch using NumPy, then with Scikit-learn. Evaluate using MSE/R^2 and Accuracy/Precision/Recall.	Implement and evaluate fundamental regression and classification models.

10	Decision Trees, Random Forests, Gradient Boosting Machines (XGBoost)	Develop a Random Forest model to predict customer churn. Tune hyperparameters using GridSearchCV. Analyse feature importance.	Build and tune powerful ensemble models for improved predictive accuracy.
11	Support Vector Machines (SVMs), K-Nearest Neighbors (KNN), Naive Bayes	Use SVMs for image classification. Compare KNN and Naive Bayes for a text classification task.	Apply kernel-based, instance-based, and probabilistic algorithms.
12	**Mid-Module Hackathon:** End-to-end classification/regression challenge.	Compete to build the most accurate predictive model on an unseen dataset. Present findings and methodology.	Consolidate learning by applying multiple algorithms to a real-world problem.
13	Unsupervised Learning: K-Means Clustering, Hierarchical Clustering, DBSCAN	Segment customers into distinct groups using K-Means. Compare clustering algorithms on datasets with different shapes.	Discover hidden patterns and structures in data using clustering algorithms.
14	Dimensionality Reduction: Principal Component Analysis (PCA), t-SNE	Apply PCA on a high-dimensional dataset. Use t-SNE for visualisation of high-dimensional data.	Mitigate the curse of dimensionality and improve model performance/visualisation.
15	Model Evaluation & Validation: Cross-Validation, ROC/AUC, F1-Score, Bias-Variance Tradeoff	Deep dive into model metrics. Implement k-fold cross-validation. Plot ROC curves and analyse the confusion matrix.	Rigorously evaluate and validate machine learning models to avoid overfitting.
16	Intro to MLOps: Model Serialization, Experiment Tracking (MLflow), API deployment (Flask)	Save a trained model. Log experiments, parameters, and metrics with MLflow. Build a simple web API for predictions.	Understand the model lifecycle. Deploy a model as a service. Track experiments.

This module provides a comprehensive understanding of classical machine learning techniques for predictive analytics and pattern recognition, covering both supervised and unsupervised learning.

- **Topics Covered:**
  - **Supervised Learning Algorithms:**
    - **Linear Regression:** Simple and multiple linear regression, assumptions, interpretation of coefficients.
    - **Logistic Regression:** Binary and multi-class classification, sigmoid function, decision boundary.
    - **k-Nearest Neighbors (k-NN):** Distance metrics, choosing `k`.
    - **Naïve Bayes:** Bayes' theorem, different Naïve Bayes classifiers (Gaussian, Multinomial).
  - **Decision Trees & Pruning:**

        - ID3, C4.5, CART algorithms.
        - Information gain, Gini impurity.
        - Overfitting in decision trees, pre-pruning and post-pruning techniques.
    - **Ensemble Methods:**
        - **Bagging:** Random Forest (feature randomness, out-of-bag error).
        - **Boosting:** AdaBoost, Gradient Boosting Machines (GBM).
        - **XGBoost & LightGBM:** Advanced gradient boosting frameworks, their advantages and parameters.
    - **Support Vector Machines (SVM):** Linear and non-linear SVM, kernels (RBF, polynomial), C parameter, gamma parameter.
    - **Model Evaluation & Selection:**
        - **Bias-Variance Trade-off:** Understanding underfitting and overfitting.
        - Performance metrics for classification (Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC-AUC curve).
        - Performance metrics for regression (MSE, RMSE, MAE, R^2).
        - **Hyper-parameter Tuning:** Grid Search, Random Search, Bayesian Optimization (using Optuna).
        - **Model Workflow:** Train-test split, k-fold Cross-Validation (CV), building robust machine learning pipelines.
    - **Model Interpretability:**
        - Feature importance (e.g., from tree-based models).
        - Partial-dependence plots (PDPs).
        - Introduction to SHAP and LIME (covered in more detail in Module 9).
    - **Unsupervised Learning:**
        - **Clustering:** k-means, DBSCAN, Hierarchical clustering (agglomerative, divisive).
        - **Dimensionality Reduction:** Principal Component Analysis (PCA), t-Distributed Stochastic Neighbor Embedding (t-SNE) for visualization.
- **Tools:** scikit-learn 1.4, XGBoost, LightGBM, MLflow tracking, Optuna, Yellowbrick.
- **Case Studies:**
    - **Bank Loan Default Prediction (Open RBI dataset):** A classification problem where students build models to predict loan defaults. Focus on handling class imbalance (as defaults are rare) and evaluating models using appropriate metrics like ROC-AUC.
    - **Customer Segmentation (Telco Churn):** Apply clustering algorithms (k-means, DBSCAN) and dimensionality reduction (PCA, t-SNE) to segment telecom customers based on their behavior, aiming to identify segments prone to churn. Visualization of clusters will be a key component.
- **Real-life Examples:**
    - **LIC's Underwriting ML Pipeline:** Discussion on how large insurance companies like LIC use ML models for risk assessment and underwriting, automating decisions and improving efficiency.
    - **Zomato's Delivery-Time ETA Model:** How food delivery platforms use ML to predict estimated time of arrival (ETA) for orders, considering factors like traffic, restaurant preparation time, and delivery distance.
- **Hands-on Activities:**
    - **Weekly Lab Notebooks:** For each algorithm, students will first implement a simplified version from scratch (e.g., simple linear regression, k-means) to

understand the underlying math, then use `scikit-learn` for full implementation on various datasets.
  - **Kaggle-style Mini-Competition:** Participate in a mini-competition on a classification or regression task, aiming for an F1 score > 0.84 (or similar target) to earn a badge, fostering competitive and practical skill development.
  - **Experiment-Tracking Assignment in MLflow:** Log 5 different model runs (e.g., varying hyper-parameters or algorithms) for a given problem, tracking metrics, parameters, and artifacts using MLflow.

3.4 Module 4: Deep Learning & Generative AI (Weeks 17–24, 240 Hours)

- **Indicative Software/Tools:** PyTorch 2.x, Hugging Face (Transformers, Datasets), OpenCV, Docker, Kubernetes (concepts), Gradio.

Weekly Breakdown:

Week	Lecture Themes	Lab Projects / Assignments	Key Learning Outcomes
17	Intro to Neural Networks, Backpropagation, PyTorch Tensors, Building DL models in PyTorch	Build a neural network from scratch using PyTorch to classify the MNIST dataset.	Understand the architecture and training of a neural network. Use a modern DL framework.
18	Convolutional Neural Networks (CNNs) for Computer Vision	Build a CNN to classify images from the CIFAR-10 dataset. Implement data augmentation to improve robustness.	Apply CNNs to solve image recognition tasks.
19	Transfer Learning with Pre-trained Models (e.g., ResNet, VGG)	Fine-tune a pre-trained model for a custom image classification task (e.g., crop disease detection).	Leverage pre-trained models to achieve high accuracy on small datasets.
20	Recurrent Neural Networks (RNNs) & LSTMs for Sequential Data	Develop an LSTM-based model for sentiment analysis on movie reviews.	Model sequential data for tasks in Natural Language Processing (NLP).
21	The Transformer Architecture, Attention Mechanism, Intro to BERT	Use a pre-trained Transformer (e.g., BERT) from Hugging Face for a text classification task. Visualise attention heads.	Grasp the fundamentals of the Transformer architecture, the foundation of modern GenAI.
22	Generative AI: Prompt Engineering, LLMs, Intro to Diffusion Models	Experiment with prompt engineering techniques. Build a simple question-answering app using an LLM API. Create a simple UI with Gradio.	Interact with and build applications on top of large language models.

23	MLOps for Deep Learning: Dockerization, CI/CD Concepts, Cloud Deployment (intro)	Containerize the Flask API using Docker. Discuss concepts of deploying models at scale using Kubernetes/cloud services.	Prepare models for robust, scalable production environments.
24	**Capstone Project Showcase & Final Evaluation**	Present the final capstone project to a panel of mentors and industry experts.	Demonstrate mastery of the entire AI/DS pipeline by delivering a comprehensive project.

This module delves into neural networks, advanced AI topics, deep learning architectures, and the foundations of generative AI, with a strong emphasis on practical implementation.

- **Topics Covered:**
  - **Neural Network Anatomy:** Perceptrons, multi-layer perceptrons (MLPs), activation functions (ReLU, Sigmoid, Tanh, Softmax), loss functions (MSE, Cross-Entropy).
  - **Optimizers:** Stochastic Gradient Descent (SGD), Momentum, AdaGrad, RMSprop, Adam, AdamW. Learning rate schedules.
  - **Convolutional Neural Networks (CNNs):**
    - Convolutional layers, pooling layers (max, average).
    - Architectures: LeNet, AlexNet, VGG, Inception, **ResNet**, **EfficientNet**.
    - Data augmentation techniques for images.
  - **Recurrent Neural Networks (RNNs):**
    - Basic RNNs for sequence data.
    - Challenges: Vanishing/exploding gradients.
    - **Gated Recurrent Units (GRUs)** and **Long Short-Term Memory (LSTM)** networks.
  - **Attention Mechanisms & Transformers:**
    - Self-attention mechanism.
    - Encoder-Decoder architecture.
    - Introduction to the **Transformer architecture** (multi-head attention, positional encoding).
  - **Pre-training vs. Fine-tuning:** Strategies for leveraging pre-trained models on new datasets.
  - **Generative Models Introduction:**
    - **Auto-encoders (AEs):** Encoder-decoder for dimensionality reduction and data reconstruction.
    - **Variational Auto-encoders (VAEs):** Probabilistic approach to generative modeling.
    - **Generative Adversarial Networks (GANs):** Generator-discriminator architecture, training challenges (mode collapse).
  - **Prompt Engineering Strategies:**
    - Zero-shot, few-shot prompting.
    - Chain-of-thought, Tree-of-thought prompting.
    - Role-playing and persona-based prompting.
  - **Transfer Learning & Freezing Layers:** Techniques for efficient model adaptation using pre-trained weights.

- **Deployment Formats:** Saving and loading models in formats like `SavedModel` (TensorFlow), ONNX, and TensorRT for optimized inference.
- **Tools:** TensorFlow 2.16, KerasCV/NLP, PyTorch 2 (optional lab for comparative understanding), Hugging Face Transformers, Weights & Biases (for experiment tracking), Google Colab GPU.
- **Case Studies:**
    - **Malaria Cell-Image Classifier (CNN):** Build and train a CNN to classify malaria-infected and uninfected cells from microscopic images, focusing on image preprocessing, data augmentation, and model evaluation for medical applications.
    - **Hindi News Sentiment:** Fine-tune a pre-trained Transformer model like DistilBERT for sentiment analysis on a dataset of Hindi news articles, addressing challenges of low-resource languages and transfer learning.
    - **MNIST → Anime Face GAN:** An introductory project to generative models where students train a simple GAN to generate anime-style faces from MNIST digits, exploring the concepts of generator, discriminator, and mode collapse.
- **Real-life Examples:**
    - **MyGov's Bhashini LLM Initiatives:** Discussion on government initiatives in India to develop large language models for various Indian languages, highlighting the importance of NLP for linguistic diversity.
    - **Dream11's Real-time Image Moderation with EfficientDet:** How platforms like Dream11 use advanced computer vision models (e.g., EfficientDet) for real-time content moderation of user-uploaded images, ensuring compliance and safety.
- **Hands-on Activities:**
    - **ResNet-34 Training on Malaria Dataset:** Implement and train a ResNet-34 model on the Malaria dataset, aiming for a validation accuracy of ≥ 94%. Focus on data loading, augmentation, training loops, and evaluation.
    - **Hindi DistilBERT Fine-tuning for Sentiment:** Fine-tune a pre-trained DistilBERT model on a Hindi sentiment dataset, achieving an F1 score of ≥ 0.80. This involves using the Hugging Face Transformers library.
    - **GAN Playground:** Experiment with a simple GAN architecture, training it to generate 50 synthetic images and analyzing phenomena like mode collapse during training.

4. Pedagogy & Daily Schedule

The pedagogical approach is intensely hands-on, following a **"30% Theory, 70% Practical"** model.

- **Theory Sessions (Lectures/Mentoring):** Morning sessions introduce new concepts, algorithms, and theoretical underpinnings.
- **Guided Labs:** Post-lecture sessions where students implement the concepts learned under the direct supervision of mentors using Industry case study.
- **Project Studio:** Dedicated time for students to work on their weekly assignments and the project, fostering collaboration and independent problem-solving.
- **Peer Learning:** Students are encouraged to perform code reviews for each other with **legacy code maintenance**, fostering a collaborative environment and improving code quality.

- **Mentoring:** One-on-one and group mentoring sessions are integrated to resolve doubts, review code, and provide career guidance.

Exploring and Maintaining legacy code, here are some websites and resources where you can find legacy code projects:

- **GitHub (https://github.com/)** - Search for old, unmaintained repositories. Keywords like "legacy code" or "deprecated projects" can help you find suitable repositories.
- **SourceForge (https://sourceforge.net/)** - An older platform with many legacy projects that are no longer actively maintained.
- **OldCode (https://oldcode.net/)** - A site dedicated to hosting legacy codebases for learning and practice purposes.
- **CodeProject (https://www.codeproject.com/)** - Offers articles and code samples, including older projects.
- **Coding Dojo Legacy Code Katas (https://codingdojo.org/kata/)** - Provides coding exercises specifically designed around legacy code scenarios.

Typical Daily Timetable (Monday - Friday):

Time Slot	**Activity**	**Duration**	**Focus**
09:30 - 11:30	Lecture Session	2.0 hrs	Core Concept Delivery (Theory)
11:30 - 11:45	Break	0.15 hrs	-
11:45 - 13:15	Guided Lab Session	1.30 hrs	Hands-on Implementation (Practical) on industry projects
13:15 - 14:00	Lunch Break	0.45 hr	-
14:00 - 15:00	Project Studio / Mentoring	1.00 hrs	Application & Problem Solving (Practical)/ /**Exposure to Freelancing Projects/ Projects on Kaggle**
15:00 - 15:30	Mentor Huddle / Reflection	0.30 hrs	Discussion
Total		**5.0 hrs**	**Intensive Learning**

5. Assessment Methodology

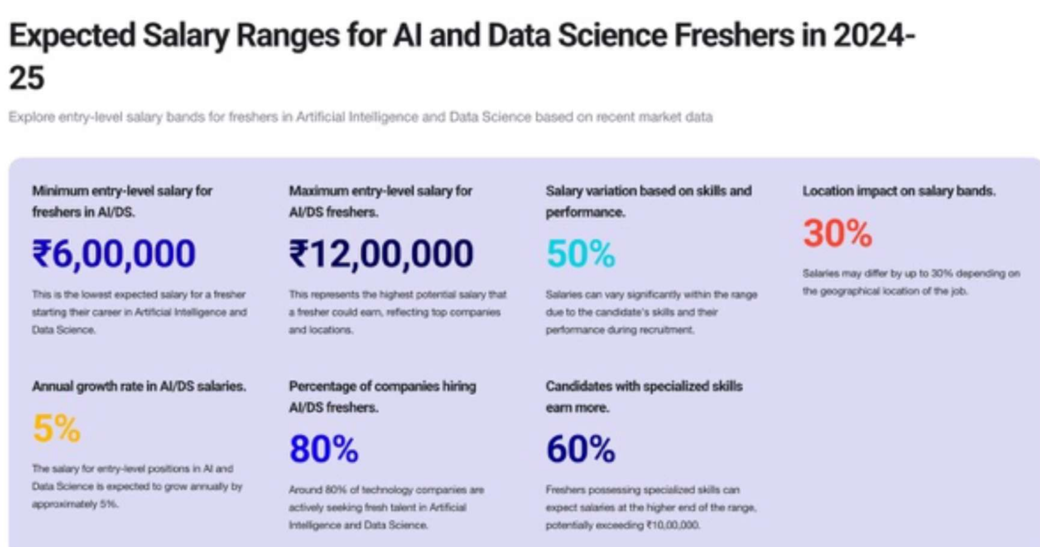
- **Weekly Quizzes (10%):** Online MCQs to test theoretical understanding.
- **Lab Assignments (20%):** Practical assignments evaluated using rubrics that assess code quality, correctness, documentation, and efficiency.

- **Real Industry Projects (70%):** A competitive event to test problem-solving and application skills.

Capstone Project Evaluation Rubric:

Criteria	Weight	Description
Problem Definition & Innovation	20%	Clarity of the problem statement, novelty of the approach, and real-world relevance.
Data Engineering & EDA	20%	Quality of data collection, cleaning, preprocessing, and exploratory data analysis.
Modeling & Experimentation	30%	Appropriateness of model choice, rigor of experimentation, and depth of evaluation.
Deployment & Presentation	20%	Successful deployment of a working prototype (e.g., API, web app) and clarity of the final presentation.
Code Quality & Documentation	10%	Readability, modularity, and version control practices; quality of the final report.

6. Career Pathways & Industry Alignment



6.1 Entry-Level Roles

Graduates will be prepared for a variety of high-demand entry-level roles:

- Data Analyst / Business Analyst

- Junior Data Scientist
- Machine Learning Engineer (Intern/Junior)
- AI Application Developer
- Data Quality Assurance / AI Tester
- MLOps Support Engineer
- Analytics Engineer

6.2 Expected Salary Bands

Based on 2024-25 data from sources like Naukri, Glassdoor, and Michael Page's India Salary Guide, a fresher with these specialised skills can expect:

- **Entry-Level Salary Range:** ₹6,00,000 to ₹12,00,000 per annum, depending on the company, location, and candidate's performance.

6.3 Vertical Career Options

With 2-3 years of experience, graduates can specialise in advanced, higher-paying verticals:

- Computer Vision (CV) Specialist
- Natural Language Processing (NLP) Engineer
- Generative AI Prompt Engineer / Application Developer
- AI/ML Product Manager
- Data Analytics Consultant

7. Alignment with International Certifications & Standards

This diploma is designed not only for job readiness but also to provide a strong foundation for globally recognized professional certifications. The curriculum intentionally covers the core competencies tested by major cloud and AI technology providers. This alignment ensures that our graduates' skills are benchmarked against international standards, enhancing their mobility and career prospects worldwide.

Mapping of Diploma Modules to Certification Domains:

Diploma Module	Key Skills Covered	Aligns with Knowledge Areas of...
Module 1 & 2	Python, Data Handling, EDA, SQL, Statistics	**Microsoft (DP-100):** Manage data assets; **AWS (MLS-C01):** Exploratory Data Analysis; **Google (PML):** Frame ML problems, Engineer data.
Module 3	Classical ML (Regression, Classification, Clustering), Model Evaluation, MLOps Intro (Flask, MLflow)	**Microsoft (DP-100):** Train, evaluate, and deploy models; **AWS (MLS-C01):** Modeling; **Google (PML):** Build ML models.
Module 4	Deep Learning (PyTorch, CNN, RNN), GenAI (Transformers, LLMs), Advanced MLOps (Docker)	**Microsoft (AI-102):** Implement computer vision & NLP solutions; **AWS (MLS-C01):** Modeling; **Google (PML):** ML Operations.

8. Pathway to International Certifications

Upon successful completion of the diploma, students will have acquired the fundamental knowledge and practical skills required to confidently attempt several prestigious international certification exams. While the diploma provides the core preparation, candidates are advised to undertake platform-specific revision and practice tests before appearing for the exams.

8.1 Foundational Level Certifications

- **Microsoft Certified: Azure AI Fundamentals (AI-900):** Validates foundational knowledge of machine learning and AI concepts and how they are implemented on Microsoft Azure. Our curriculum, especially Modules 1 and 3, covers all core AI workloads discussed in this exam.

8.2 Associate & Specialty Level Certifications

- **Microsoft Certified: Azure Data Scientist Associate (DP-100):** This certification is a direct fit. Our diploma extensively covers its key domains: designing and preparing a machine learning solution, training models, and deploying and managing solutions using Azure Machine Learning. Modules 2, 3, and 4 provide a solid base for this exam.
- **AWS Certified Machine Learning – Specialty (MLS-C01):** This is a highly respected certification. Our programme's focus on Data Engineering (Module 2), Modeling with a variety of algorithms (Module 3 & 4), and MLOps concepts provides strong preparation for the key domains of this exam.
- **Google Cloud Certified - Professional Machine Learning Engineer:** This certification focuses on building and productionizing ML models on Google Cloud. Our end-to-end project-based approach, from data engineering to MLOps (Docker, Kubernetes concepts), aligns perfectly with the skills required for this advanced certification.

9. Learning Centre Infrastructure Requirements

9.1 Hardware Specifications (per workstation)

- **CPU:** Intel Core i7 / AMD Ryzen 7 (or equivalent)
- **RAM:** 16 GB DDR4 (32 GB recommended)
- **GPU:** NVIDIA GeForce RTX 3060 (12 GB) or higher (essential for Module 4)
- **Storage:** 512 GB NVMe SSD
- **Display:** Dual 24-inch monitors

9.2 Software Stack

- **Operating System:** Ubuntu 22.04 LTS (or Windows 11 with WSL2)
- **IDEs:** VS Code, JupyterLab
- **Core Libraries:** Python 3.11+, Anaconda/Miniconda
- **Containers:** Docker Desktop
- **Collaboration:** Git, GitHub

9.3 Centre Facilities

- **Network:** High-speed internet (100+ Mbps) with low latency.
- **Power:** 100% power backup (UPS + Generator) for all workstations and servers.
- **Audio-Visual:** High-definition projector, whiteboard, and a quality audio system.
- **Safety:** Adherence to all fire safety and electrical safety norms.

9.4 Mentoring Staff

- **Mentor-to-Student Ratio:** 1:15 for optimal personal attention.
- **Lead Mentor Qualification:** MTech/PhD in CS/AI or BE with 5+ years of industry experience.
- **Lab Mentor Qualification:** BE/MCA with 2+ years of hands-on experience in Python and ML frameworks.

10. Draft Letter to Partner Centres

[On Official MKCL Letterhead]

Date: 07 July 2025

To,
The Director/Principal,
[Partner Centre Name],
[Partner Centre Address]

Subject: Invitation to Partner for Hosting the "Diploma in AI & Data Science" Programme

Dear Sir/Madam,
Greetings from Maharashtra Knowledge Corporation Ltd. (MKCL)!
As you know, India is at the forefront of a technological revolution driven by Artificial Intelligence and Data Science. To meet the immense industry demand for skilled professionals, MKCL is launching a flagship six-month, full-time “Diploma in AI & Data Science”. This programme is designed to equip engineering students with the practical, hands-on skills required to build a successful career in this exciting field.
We believe that your esteemed institution, with its commitment to quality education, would be an ideal partner to host this programme. By partnering with us, you will be at the forefront of AI education in your region, offering a high-value programme to students and strengthening your institution's brand.
We are looking for partners who can provide the necessary infrastructure to ensure a high-quality learning experience. The key requirements are enclosed in the checklist below.

Infrastructure & Facility Checklist:

1. **Dedicated Classroom:** A well-ventilated classroom with a seating capacity for 60 students.
2. **Computer Lab:** A lab with at least 20 high-performance workstations meeting the specified configuration (Intel i7/Ryzen 7, 16GB+ RAM, NVIDIA RTX 3060+ GPU, 512GB SSD). Cost per system will be Rs 1 lac
3. **High-Speed Internet:** A reliable, high-speed internet connection (>100 Mbps).
4. **Power Backup:** Uninterrupted Power Supply (UPS) for all workstations and critical equipment.
5. **AV Equipment:** Projector, screen, and audio system for effective teaching.
6. **Qualified Staff:** Willingness to nominate faculty/staff for our "Train the Mentor" programme.

We will provide the complete curriculum, learning materials, assessment platform and branding support.
We kindly request you to review this proposal and confirm your interest by __ 2025. Our team will then schedule a visit to your campus to discuss the partnership in further detail.
We are confident that this collaboration will be mutually beneficial and will play a pivotal role in shaping the future of tech talent in India.
Thank you for your time and consideration.
Sincerely,
[Name of Authorised Signatory]

Head, Academic Programmes
Maharashtra Knowledge Corporation Ltd. (MKCL)
Pune, India

11. Admission & Fee Structure

- **Selection Process:**
  1. **Online Application:** Submission of academic details and a statement of purpose.
  2. **Aptitude Test:** A 60-minute online test covering Logical Reasoning, Quantitative Ability, and Programming Fundamentals (Python).
  3. **Admission:** Based on test scores, on a first-come, first-served basis for qualified candidates.
- **Fee Structure:**
  - **Total Programme Fee:** ₹1,00,000 + GST.
  - **Early Bird Discount:** 10% discount for admissions confirmed 30 days before the batch start date.
  - **Need-Based Scholarship:** Up to 25% scholarship available for a limited number of meritorious students from economically weaker sections.
  - **Financing Options:** Tie-ups with leading financial institutions for no-cost EMI options.

12. Academic Calendar

- **Monsoon Batch (July - December):**
  - Admissions Open: April 1st
  - Programme Start Date: Second Monday of July
- **Spring Batch (January - June):**
  - Admissions Open: October 1st
  - Programme Start Date: Second Monday of January

13. Quality Assurance & Governance

- **Mentor KPIs:** Performance measured based on student feedback scores, student project outcomes, and course completion rates.
- **Student Feedback Loops:** Anonymous feedback collected at the end of each module to continuously improve content and delivery.
- **Curriculum Review Committee:** A committee of academic and industry experts will meet every six months to update the syllabus, ensuring its relevance to the latest industry trends.
- **ISO 9001 Alignment:** All processes will be documented and audited to align with ISO 9001:2015 standards.

14. Risk Analysis & Mitigation

Risk	Likelihood	Impact	Mitigation Strategy
Faculty/Mentor Shortage	Medium	High	Implement a robust "Train the Mentor" programme. Offer competitive remuneration. Partner with industry experts for guest lectures.
High GPU/Hardware Costs	High	Medium	Explore cloud-based GPU solutions (e.g., Google Colab Pro, AWS SageMaker) as a

			supplement. Negotiate bulk purchase discounts.
Rapidly Changing Technology	High	High	The Curriculum Review Committee will ensure agile updates to modules every six months.
Ethical Misuse of AI	Medium	High	Integrate a mandatory "AI Ethics and Responsible AI" thread throughout the curriculum, with specific case studies and a capstone project evaluation criterion on ethical considerations.
Low Student Enrolment	Low	High	Strong digital marketing, free workshops in engineering colleges, and highlighting the international certification pathway.

Appendices

A.1 Glossary of AI Terms

- **Algorithm:** A set of rules or instructions given to an AI to help it learn.
- **Backpropagation:** A method used in training neural networks to update network weights.
- **CNN (Convolutional Neural Network):** A type of deep learning model for processing grid-like data, such as an image.
- **Docker:** A platform for developing, shipping, and running applications in containers.
- **Generative AI:** AI models that can create new, original content.
- **LLM (Large Language Model):** An AI model trained on vast amounts of text data.
- **MLOps (Machine Learning Operations):** Practices for deploying and maintaining ML models in production reliably.
- **Overfitting:** A modeling error where a model learns training data noise and fails to generalize.
- **PyTorch:** An open-source machine learning library for applications like CV and NLP.
- **Responsible AI:** An approach to developing and deploying AI systems that are safe, trustworthy, and ethical.
- **Transfer Learning:** Reusing a pre-trained model as the starting point for a second task.

A.2 Suggested Reading List & MOOCs

Books:

- *Python for Data Analysis* by Wes McKinney
- *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow* by Aurélien Géron
- *Deep Learning with PyTorch* by Eli Stevens, et al.
- *Designing Data-Intensive Applications* by Martin Kleppmann

MOOCs (for pre-learning/reference):

- *Machine Learning* by Andrew Ng (Coursera)

- *Deep Learning Specialization* by deeplearning.ai (Coursera)
- *Python for Everybody* by University of Michigan (Coursera)

A.3 Example Capstone Project Briefs

1. **AI-Powered Crop Disease Detection:** Build a web app that identifies crop diseases from an uploaded leaf image using CNNs and Transfer Learning.
2. **Bilingual Customer Support Chatbot:** Develop a conversational AI chatbot for e-commerce that can handle queries in both Hindi (Roman script) and English using LLMs.
3. **Credit Risk Model for FinTech:** Analyse a loan dataset to build a model that classifies applicants as "good" or "bad" credit risks using XGBoost.
4. **Generative AI for Marketing Copy:** Create a tool that takes a product description and generates engaging marketing copy (e.g., social media posts, ad headlines) using prompt engineering with a powerful LLM.

1. Core Faculty Roles & Typical Qualifications

Role	Typical Background & Qualifications	Modules they can comfortably own	Why they fit
Foundations & Data Engineering Lecturer	• M.Sc./Ph.D. in Computer Science / Data Science • 5 + yrs teaching intro Python, statistics & databases • Industry stint in data engineering or analytics (pandas/Spark/SQL at scale) • Active GitHub portfolio, OSS contributions	**1 & 2** – Foundations of AI/Data Sci & Data Handling/Visualisation	Same tool-chain (Python, NumPy/Pandas, SQL, Spark, Tableau). Pedagogically sensible to keep early-stage momentum with one familiar instructor.
Machine Learning Scientist	• Ph.D. (or strong M.Tech + peer-reviewed publications) in Machine Learning / Applied Statistics • 5–8 yrs building models in scikit-learn/XGBoost • Kaggle or industry track record (credit-risk, churn, etc.)	**3** – Classical ML	Requires depth in supervised/unsupervised theory, but not heavy neural-network focus.
Deep-Learning &	• Ph.D. or equivalent research output in	**4, 5, 6** – Deep Learning, NLP, Computer Vision	Shared foundations: back-prop, transformers, GPUs.

Role	Typical Background & Qualifications	Modules they can comfortably own	Why they fit
Gen AI Specialist	Deep Learning / Computer Vision / NLP • Extensive PyTorch or TensorFlow engineering in production (ideally LLM/RLHF exposure) • Publicly released models or papers		One seasoned DL engineer can span all three, bringing coherent view of modern foundation-model stack.
Software-Engineering & MLOps Coach	• B.E./M.Tech in Software Eng. + 8 yrs enterprise experience • Certified Kubernetes Administrator (CKA) or AWS/GCP DevOps Pro • Contributor to open-source DevOps / CI-CD tooling	**7 & 8** – AI for Software Eng + MLOps & Deployment	Strong overlap: static code analysis, Docker/K8s, CI-CD, feature stores.
AI Ethics & HCD Specialist	• Master's in HCI, Design, or Tech-Policy • Research/consulting on AI fairness & accessibility • Member of standards body or published in FAT*/CHI venues	**9** – Ethics & Human-Centred Design	Requires cross-disciplinary skill set rarely found in purely technical faculty.
Capstone Director & Project Mentors	• 10 + yrs leading data/AI projects in industry • Experience supervising student or intern teams • Comfortable with legacy-modernisation programmes	**10** – Capstone Project	Coordinates domain mentors, runs weekly design reviews, liaises with industry client.

Total distinct full-time faculty required: ***6*** *(rows above).*

---

2. Supporting Roles

Role	Typical Profile	Load
Teaching Assistants (TAs)	Recent graduates (M.Sc./M.Tech) who excelled in prior cohort or top-tier universities; strong coding chops; each TA paired with 1 – 2 faculty.	3 – 4 full-time equivalents
Adjunct Industry Speakers	Domain experts from fintech, health-tech, e-commerce to give 2-hour masterclasses or case-study walk-throughs.	1 per module, guest basis
UX/Design Studio Coach	Senior product designer familiar with WCAG & service design.	Part-time during Module 9 & Capstone

---

3. How Module – Faculty Pairing Works

1. **Modules 1 & 2** share the same datasets, notebooks and tooling; students benefit from continuity.
2. **Modules 4** (DL, NLP, CV) all rely on the transformer ecosystem; one expert can contextualise trade-offs (e.g., Vision Transformers vs BERT).
3. **Capstone** leverages *all* faculty as domain mentors, but the Project Director owns grading rubrics and stakeholder communication coaching.

---

4. Recommended Hiring Criteria & Evidence

Competency	Evidence to seek
Pedagogical skill	Prior syllabus design, student feedback > 4/5, recorded lectures or MOOCs
Research/Industry impact	Google Scholar citations, GitHub stars, patents, Kaggle medals, conference talks
Collaborative mindset	History of co-authoring papers/repos, mentoring interns, contributing to OSS communities
Diversity, Equity & Inclusion	Experience embedding ethical tooling (SHAP, Aequitas) or accessibility guidelines in past work

---

5. Final Notes on Faculty Load Sharing

- **Yes, multiple modules can—and often should—be taught by a single faculty** when:
    - The knowledge graph overlaps strongly (e.g., classical ML → deep learning transition).
    - Maintaining narrative coherence helps learners see “the bigger picture”.
- **Limit per-faculty contact hours** to ~180–200 over 48 weeks to avoid burnout; augment with TAs for labs and doubt-clearing.

- **Have at least one "swing" adjunct** (e.g., cloud-ML architect) who can step in for guest sessions or cover spill-over if a primary instructor is unavailable.

By structuring the teaching team in this way, the programme remains **cost-efficient, intellectually coherent, and attractive to learners who value instructors with both scholarly depth and real-world delivery experience.**

IT Infrastructure requirements:

1. **Dedicated classroom** for 60 learners (AC, ergonomic seating, HD projector, whiteboard).
2. **Computer lab** with *minimum* 60 workstations:
   - Intel Core i7-12700F / AMD Ryzen 7 5800X equivalent
   - 16 GB RAM (32 GB preferred)
   - NVIDIA RTX 3060 6 GB GPU
   - 512 GB NVMe SSD
     (street-price ₹ 60,000 – 70,000 per unit)
3. 100-Mbps symmetrical internet with wired back-up line.
4. 30 kVA UPS + generator to support a 3-hour power-outage window.
5. Faculty release time (1 Lead, 2 Lab mentors) and 1 System Admin.
6. Willingness to maintain an **AI compute pool**: either
   - one on-prem multi-GPU server **or**
   - ₹ 12,000 cloud-GPU credit per student (USD 150).

Tentative infrastructure budget (per centre)

CAPEX

Item	Qty	Unit est. (₹)	Sub-total (₹)	Notes
High-spec Desktop (RTX 3050)	30	70,000	**21,00,000**	Prices from Google
Core/edge switch + Wi-Fi 6 APs	1 set	1,00,000	1,00,000	Managed PoE switch, dual WAN
30 kVA Online UPS + batteries	1	2,50,000	2,50,000	30-min full-load, plus genset hook-up
AV system (Smart TV, mic-bar)	1	1,00,000	1,00,000	Includes motorised screen
Furniture & ergonomic chairs	—	—	2,00,000	For lab + lecture room
Software & licences (Win 11, Office)	—	—	10,000	Open-source stack otherwise
Indicative capex total			**27,60,000**	(≈ ₹ 28 lakh)

Optional on-prem GPU server: 4×RTX 4090 build ≈ ₹ 9 lakh **or** single A100 80 GB card ≈ ₹ 19.2 lakh as of Nov 2025 serverbasket.com. College may instead rely on cloud-GPU vouchers.

OPEX

Item	Qty	Unit est. (₹)	Sub-total (₹)
Cloud-GPU (1,50,000 x 6)	6 months	1,50,000	~9,00,000
Learning Facilitators	2	50,000	6,00,000
Approximate Opex			**~15,00,000**

1. SUMMARY OF COST STRUCTURE

Cost Category	College	MKCL	Notes
CAPEX (One-time Setup)	₹27,60,000	₹0	College bears infra setup.
OPEX (Per 6-month Batch)	₹15,00,000	₹4,00,000	MKCL costs include curriculum, LMS, assessments, certification, QA.
Revenue (Per Batch @ ₹1,00,000 fee)	Shared	Shared	Final sharing model detailed below.

---

2. DETAILED COST SHEET (COLLEGE PERSPECTIVE)

A. Capital Expenditure (One-time Cost)

AI-DS Foundation Diploma

CAPEX Item	Qty	Unit Cost (₹)	Total (₹)	Notes
High-spec Desktops (RTX 3050)	30	70,000	**21,00,000**	For DL/GPU workloads
Network infra (Switch + Wi-Fi 6 APs)	1	1,00,000	**1,00,000**	Managed PoE switch
UPS 30 kVA + batteries	1	2,50,000	**2,50,000**	Backup for 3 hours
AV System (Smart TV, mic-bar)	1	1,00,000	**1,00,000**	Lecture delivery
Furniture (lab + class)	—	—	**2,00,000**	Ergonomic
Software licenses (minimal)	—	—	**10,000**	Mostly open-source
Total CAPEX (College)			**₹27,60,000**	One-time setup

B. Operating Expenditure (Recurring per Batch)

AI-DS Foundation Diploma

OPEX Item	Duration	Cost (₹)	Notes
Cloud GPU credits	6 months	**9,00,000**	Optional supplement to lab GPUs
Learning Facilitators (2 × 50,000 × 6 months)	6 months	**6,00,000**	Local teaching support
System Admin	Included	—	Usually absorbed by college
Electricity, maintenance, consumables	6 months	Included	Often clubbed into infra cost
Total OPEX (College)		**₹15,00,000**	Per batch

C. College Annualised Cash Flow Estimate

Assume **2 batches per year**.

Component	Amount (₹)
Total Annual OPEX (2 batches)	**30,00,000**
CAPEX amortised over 3 years	**9,20,000 per year**
Total Annual Cost (College)	**≈ 39,20,000**

---

3. MKCL COST SHEET

MKCL's recurring costs are based on:

- Curriculum development & updates
- Assessments, certification, QA
- Mentor training
- LMS/Platform usage
- Branding, marketing support

MKCL Core Costs (per batch)

Cost Head	Amount (₹)
LMS platform + hosting + data	**1,00,000**
Content licensing, updates, QA	**1,20,000**
Mentor training & onboarding	**80,000**
Central assessment & certification	**1,00,000**
Total OPEX (MKCL)	**₹4,00,000**

---

4. REVENUE MODEL (60-STUDENT BATCH)

From the document: **Programme Fee = ₹1,00,000 + GST per learner**
Assuming GST handled separately, we calculate on the base amount.

Total Revenue per Batch

60 students × ₹1,00,000 = **₹60,00,000**

---

5. RECOMMENDED REVENUE SHARING MODEL

Based on the cost structure and industry norms:

Shareholder	Percentage	Amount (₹)
College	50%	**30,00,000**
MKCL	50%	**30,00,000**
Total	100%	**60,00,000**

This distribution works because:

- College bears heavy CAPEX + OPEX (~₹19.5 lakh per batch including amortisation).
- MKCL bears academic & operational quality cost (~₹4 lakh) + strategic oversight.

---

6. PROFITABILITY ANALYSIS

A. College Profit (Per Batch)

Item	Amount (₹)
College Revenue Share	**30,00,000**
College OPEX	**15,00,000**
CAPEX Amortisation (per 6 months)	**4,60,000**
Net Surplus (College)	**≈ 10,40,000**

B. MKCL Profit (Per Batch)

Item	Amount (₹)
MKCL Revenue Share	**30,00,000**
MKCL OPEX	**4,00,000**

Item	Amount (₹)
Net Surplus (MKCL)	**≈ 26,00,000**

---

7. SENSITIVITY ANALYSIS (Batch Fill Rate)

If batch has fewer than 60 students:

Batch Size	Total Revenue	College Share	College Surplus	MKCL Share	MKCL Surplus
60	60,00,000	30,00,000	**10,40,000**	30,00,000	**26,00,000**
50	50,00,000	25,00,000	**5,40,000**	25,00,000	**21,00,000**
40	40,00,000	20,00,000	**40,000 (BE point)**	20,00,000	**16,00,000**
30	30,00,000	15,00,000	**(4,60,000) Loss**	15,00,000	**11,00,000**

Break-even for college = >**40 students**.

---

8. FINAL CONSOLIDATED COST SHEET

A. Total Cost Responsibilities

Cost Type	College	MKCL
CAPEX	**100%**	—
Teaching infrastructure OPEX	**100%**	—
Cloud/AI compute	**100%**	—
Learning platform	—	**100%**
Curriculum & updates	—	**100%**
Assessment & certification	—	**100%**
Programme governance	—	**100%**

---

9. EXECUTIVE SUMMARY

Stakeholder	Investment	Revenue	Surplus
College	High CAPEX + OPEX	₹30 lakh share	**~₹10.4 lakh**
MKCL	Low OPEX	₹30 lakh share	**~₹26 lakh**